

Ecological Impact Assessment

Millbrook Solar & Storage – Marlow Ecology Ltd, report MAR-2026-117

Liam McAllister · 12 June 2026 · MAR-2026-117

Non-technical summary

Marlow Ecology Ltd was commissioned by Weyland Energy to assess the ecological effects of the proposed Millbrook Solar & Storage scheme. Surveys between October 2025 and June 2026 covered habitats, hedgerows, breeding and wintering birds, bats, badger, great crested newt, otter and water vole. The Site is dominated by intensively managed arable land of low ecological value. The most important ecological feature is a breeding population of **skylark** (23 territories).

With the proposed mitigation – 46 skylark plots, wildflower margins, 2.4 km of new hedgerow and a 10 m buffer along Mill Brook – no significant residual effects are predicted, and the scheme delivers a biodiversity net gain of **+20.1 %** in habitat units.

1. Survey effort

Survey	Dates	Surveyor(s)
Extended Phase 1 / UKHab habitat survey	14–15 Oct 2025	R. Moss, J. Okafor-Price
Wintering bird survey (4 visits)	Nov 2025 – Feb 2026	T. Haines
Breeding bird survey (6 visits)	Apr – Jun 2026	T. Haines, R. Moss
Bat activity transects and static detectors	Apr – Jun 2026	L. Carver
Great crested newt eDNA (3 ponds)	28 Apr 2026	J. Okafor-Price
Otter and water vole (Mill Brook)	12 May 2026	L. Carver
Hedgerow survey (Hedgerows Regulations)	15 Oct 2025	R. Moss

2. Designated sites

There are no statutory designated sites within 2 km. The nearest is Ashcombe Wood Local Wildlife Site (0.9 km north-east), which is not hydrologically or functionally linked to the Site.

3. Key ecological features

3.1 Skylark

Skylark is a priority species of principal importance. 23 territories were mapped across the eleven fields, the highest densities in fields 5, 6 and 9. Skylarks avoid nesting close to tall structures, so the arrays are expected to displace most territories within the panel area. See the separate Breeding Bird Survey report MAR-2026-121.

3.2 Hedgerows

6.7 km of hedgerow, of which 1.9 km qualifies as *important* under the Hedgerows Regulations 1997. 0.6 km of low-value hedgerow will be removed for access and the BESS compound; none of it is important.

3.3 Bats

Common and soprano pipistrelle dominate (87 % of passes). Noctule and a small number of barbastelle passes were recorded along the Mill Brook corridor. No roosts will be affected; lighting will be motion-activated and directed away from hedgerows and the brook.

3.4 Great crested newt, otter, badger

All three pond eDNA samples were negative for great crested newt. Otter spraints were found at the Mill Brook bridge; the 10 m buffer protects the corridor. One outlier badger sett lies in Ashcombe hedgerow (H12); a 30 m exclusion zone will apply during construction.

4. Biodiversity net gain

Unit type	Baseline	Post-development	Change
Habitat units	297.20	356.96	+20.1 %
Hedgerow units	34.40	57.20	+66.3 %
Watercourse units	4.40	5.50	+25.0 %

The calculation uses the statutory biodiversity metric; the full calculation is in the metric workbook.

5. Mitigation and enhancement

- 46 skylark plots (4 × 4 m) on 12.8 ha of retained arable land in fields 7 and 12, with five years of monitoring
- 12.8 ha of species-rich wildflower margins along hedgerows and FP14
- 2.4 km new native species-rich hedgerow; existing hedgerows managed at ≥ 3 m
- Sheep grazing at low density under the arrays; no herbicide use except for spot treatment
- Mammal gates every 50 m in the perimeter fence
- Construction outside the bird breeding season (March–August) in fields with active territories, or after a pre-start nesting check
- Landscape and Ecological Management Plan (LEMP) for 40 years, secured by condition

6. Conclusion

With mitigation, residual effects on all ecological features are *not significant*. The scheme provides a clear and measurable biodiversity net gain well above the 10 % statutory minimum.